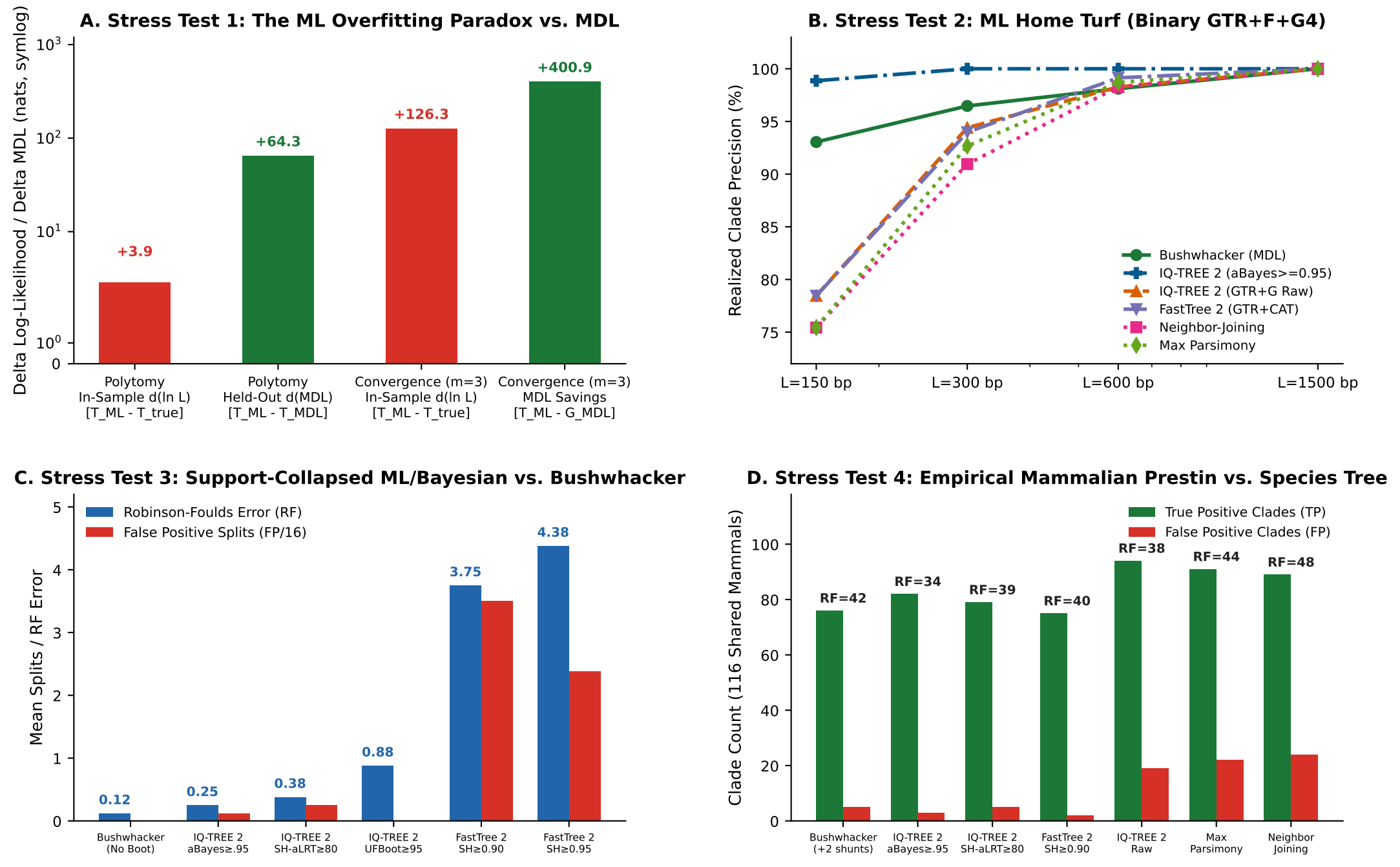
A. Stress Test 1: The ML Overfitting Paradox vs. MDL
Delta Log-Likelihood / Delta MDL (nats, symlog)
+3.9
+64.3
+126.3
+400.9
Polytomy
In-Sample d(ln L)
[T_ML - T_true]
Polytomy
Held-Out d(MDL)
[T_ML - T_MDL]
Convergence (m=3)
In-Sample d(ln L)
[T_ML - T_true]
Convergence (m=3)
MDL Savings
[T_ML - G_MDL]
B. Stress Test 2: ML Home Turf (Binary GTR+F+G4)
Realized Clade Precision (%)
L=150 bp
L=300 bp
L=600 bp
L=1500 bp
Bushwhacker (MDL)
IQ-TREE 2 (aBayes>=0.95)
IQ-TREE 2 (GTR+G Raw)
FastTree 2 (GTR+CAT)
Neighbor-Joining
Max Parsimony
C. Stress Test 3: Support-Collapsed ML/Bayesian vs. Bushwhacker
Robinson-Foulds Error (RF)
False Positive Splits (FP/16)
Mean Splits / RF Error
0.12
0.25
0.38
0.88
3.75
4.38
Bushwhacker
(No Boot)
IQ-TREE 2
aBayes≥.95
IQ-TREE 2
SH-aLRT≥80
IQ-TREE 2
UFBoot≥95
FastTree 2
SH≥0.90
FastTree 2
SH≥0.95
D. Stress Test 4: Empirical Mammalian Prestin vs. Species Tree
True Positive Clades (TP)
False Positive Clades (FP)
Clade Count (116 Shared Mammals)
RF=42
RF=34
RF=39
RF=40
RF=38
RF=44
RF=48
Bushwhacker
(+2 shunts)
IQ-TREE 2
aBayes≥.95
IQ-TREE 2
SH-aLRT≥80
FastTree 2
SH≥0.90
IQ-TREE 2
Raw
Max
Parsimony
Neighbor
Joining